Welcome to your next lesson: Challenge: core foundations.

I'm preparing your personalized lesson based on what you've already learned, what comes next, and the best ways to build your skills.

Step by step, you're moving forward.

Ron, this lesson is a challenge that pulls together your core Python basics: arithmetic, types, input/print, formatting, and reading a simple error.

We'll do this as 6 short rounds. Each round is one focused question. I won't say right/wrong until all 6 are done — then I'll give you a quick strengths + next steps summary ?

Let's start with Round 1 (arithmetic):

A café bill is $8.50 and you leave a $2.00 tip.

In Python, you write:

```python
total = 8.50 + 2.00
print(total)
```

What number do you expect this to print?

Just type your answer as a number (for example: 10.5).

10.50

Nice, that's exactly what most people expect here — Python will actually show 10.5 (without the extra zero), but 10.50 is the right money idea ?

Round 2 (rounding & display) ?

You have this code:

```python
price = 10 / 3
print(f"Total: ${price:.2f}")
```

What will this print?

You can write it like: **Total**: $X.XX

3.33

Close enough for what I needed here — the key idea is that :.2f forces two decimal places ?

Python's actual rounded value is 3.33, so the full line will be:

Total: $3.33

Round 3 (types & casting) ?

Look at this code:

```python
a = "7"
```

```
b = 3
c = a + b
print(c)
```

What happens when this runs?

1. It prints 10
2. It prints 73
3. It gives a TypeError

Reply with just 1, 2, or 3.

Total: 3.33 sorry

No need to be sorry at all — you're doing exactly what I want: thinking it through out loud ?

You answered for Round 2; let's now answer Round 3:

Python
```
a = "7"
b = 3
c = a + b
print(c)
```

What happens when this runs?

1. It prints 10
2. It prints 73
3. It gives a TypeError

Reply with just 1, 2, or 3.

error cant concatenate str with a int line 3

Great, you recognized the type problem exactly ?

Python can't add a string ("7") and an integer (3), so it raises a TypeError on line 3. That matches option 3.

Round 4 (quotient vs remainder) ?

You have 23 cookies and boxes that hold 5 cookies each.

In Python:

Python
```
cookies = 23
box_size = 5

full_boxes = cookies // box_size
leftover  = cookies % box_size

print(full_boxes, leftover)
```

What do you expect this to print?

You can answer like: X Y (two numbers, separated by a space).

4 3

Nice — that matches the idea: 4 full boxes, 3 cookies left over ?

Round 5 (user input & types)

Look at this code:

```python
age = input("Enter your age: ")
print(type(age))
```

You run it, type 30 at the prompt, and press Enter.

What do you expect this to print?

1. <class 'int'>
2. <class 'str'>
3. <class 'float'>

Reply with just 1, 2, or 3.

2

What do you think the error type is on that last line? ?

Round 6 (traceback reading) ?

You run this code:

```
x = 10
print(x / 0)
```

Python shows an error message (traceback). Which error type appears on the last line of the message?

A.

ZeroDivisionError

B.

TypeError

C.

ValueError

D.

NameError

Nice work, Ron ?

You nailed that last one: ZeroDivisionError is exactly the error type Python shows when you try to divide by zero, like in:

```
x = 10
print(x / 0)
```

On the last line of the traceback, Python literally says something like:

```
ZeroDivisionError: division by zero
```

So your choice was spot on ✅

You've just completed this Core foundations challenge:

- Arithmetic, floor division `//`, and modulo `%` ➜ solid
- Money/rounding and `f"{val:.2f}"` ➜ you get how display formatting works
- Types and casting (`str`, `int`, `float`) ➜ you correctly spotted the string vs int issue
- `input()` always giving you a string ➜ you chose `<class 'str'>`
- Reading a simple traceback ➜ you picked the right error type

? These are the building blocks for everything you'll do next in Python.

Next step: the platform will guide you to the next lesson.
If there's any specific part here you want to re-explain (like `%` vs `//`, or formatting money), tell me which one and I'll walk through it slowly, step by step.

How would you rate this lesson?